

Computer Science & IT

Discrete Mathematics



Graph Theory

Lecture No. 02



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Recap of Previous Lecture



★ Topic

Representation of graph

★ Topic

Simple graph, multi-graph and pseudo graph

★ Topic

Maximum number of edges in a simple graph with n vertices

★ Topic

Number of simple graphs possible with n labeled vertices

Topics to be Covered



Topic

Terminologies related to graphs



Topic

Different types of graphs

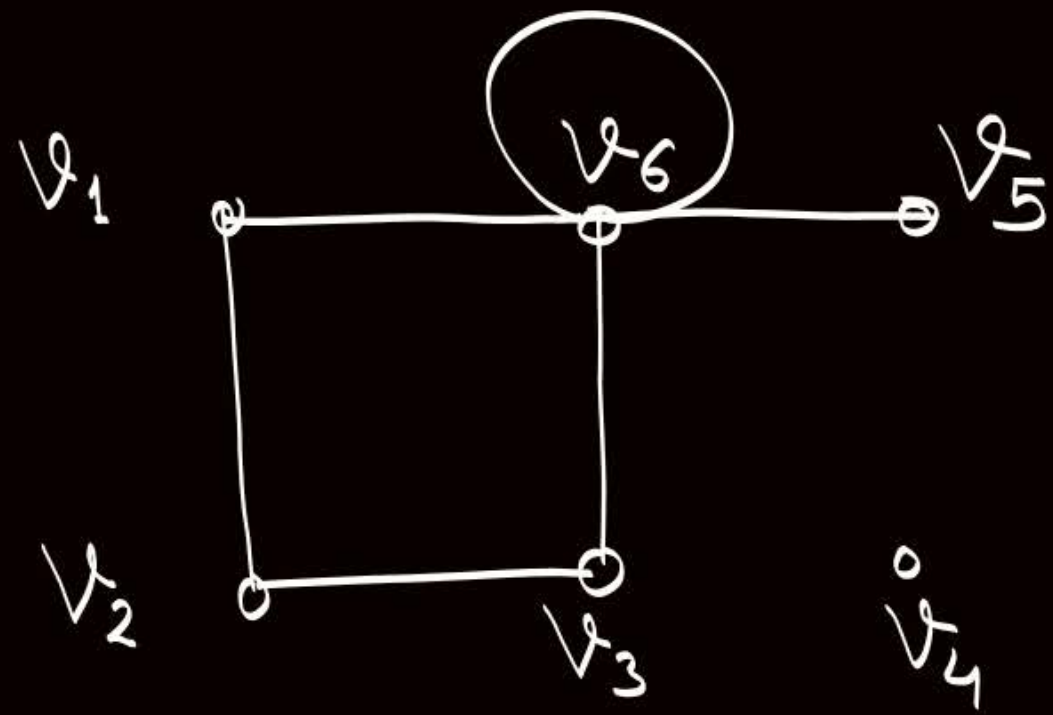




Topic : Degree of a vertex in an Undirected Graph

In an undirected graph degree of a vertex ' v ' can be denoted by $\deg(v)$ and it is defined as no. of edges incident with vertex ' v '

$$\deg(v) = \text{No. of edges incident with vertex } 'v'$$



$$\deg(v_1) = 2$$

$$\deg(v_2) = 2$$

$$\deg(v_3) = 2$$

$$\deg(v_4) = 0$$

$$\deg(v_5) = 1$$

$$\deg(v_6) = 3 + 2 = 5$$

In an undirected graph a loop is counted as two edges to obtain the degree of that vertex



Topic : Degree of a vertex in a directed Graph

In a directed graph we define two types of degrees for a vertex

- ① In-degree
- ② Out degree



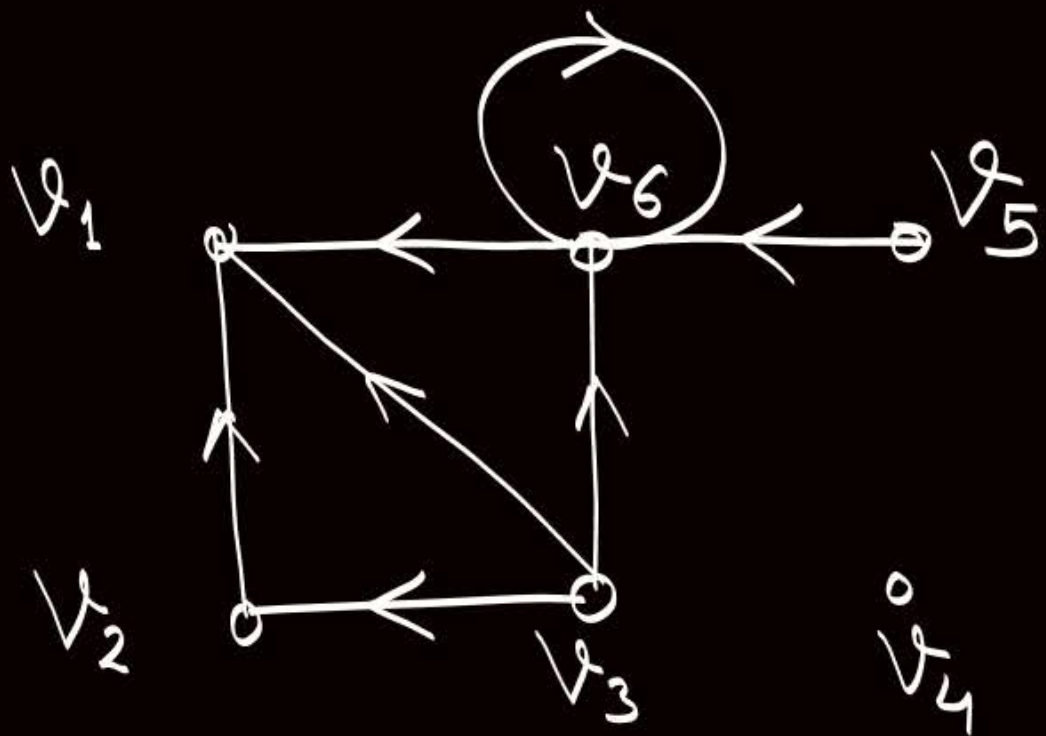
Topic : Degree of a vertex in a directed Graph

In-degree :- In-degree of a vertex ' v ' is denoted by $\deg^+(v)$ and it is defined as

$\deg^+(v)$ = No. of edges incoming to vertex v

Out-degree :- Out-degree of a vertex ' v ' is denoted by $\deg^-(v)$, and it is defined as

$\deg^-(v)$ = No. of edges out going from vertex v



In a directed graph a loop at a vertex will be counted as '1' edge to obtain the in-degree of that vertex and same loop will be counted as '1' edge to obtain out-degree of that vertex.

$$\deg^+(v_1) = 3$$

$$\deg^+(v_2) = 1$$

$$\deg^+(v_3) = 0$$

$$\deg^+(v_4) = 0$$

$$\deg^+(v_5) = 0$$

$$\deg^+(v_6) = \overset{\text{simple edges}}{2} + \overset{\text{w.r.t loop}}{1} = 3$$

$$\deg^-(v_1) = 0$$

$$\deg^-(v_2) = 1$$

$$\deg^-(v_3) = 3$$

$$\deg^-(v_4) = 0$$

$$\deg^-(v_5) = 1$$

$$\deg^-(v_6) = 1 + 1 = 2$$

$$\Sigma = 3$$

$$\Sigma = 2$$

$$\Sigma = 3$$

$$\Sigma = 0$$

$$\Sigma = 1$$

$$\Sigma = 5$$

The summation of in-degree and out-degree of a vertex 'v' will give us the overall deg of vertex 'v'

$$\text{i.e., } \deg(v) = \deg^+(v) + \deg^-(v)$$

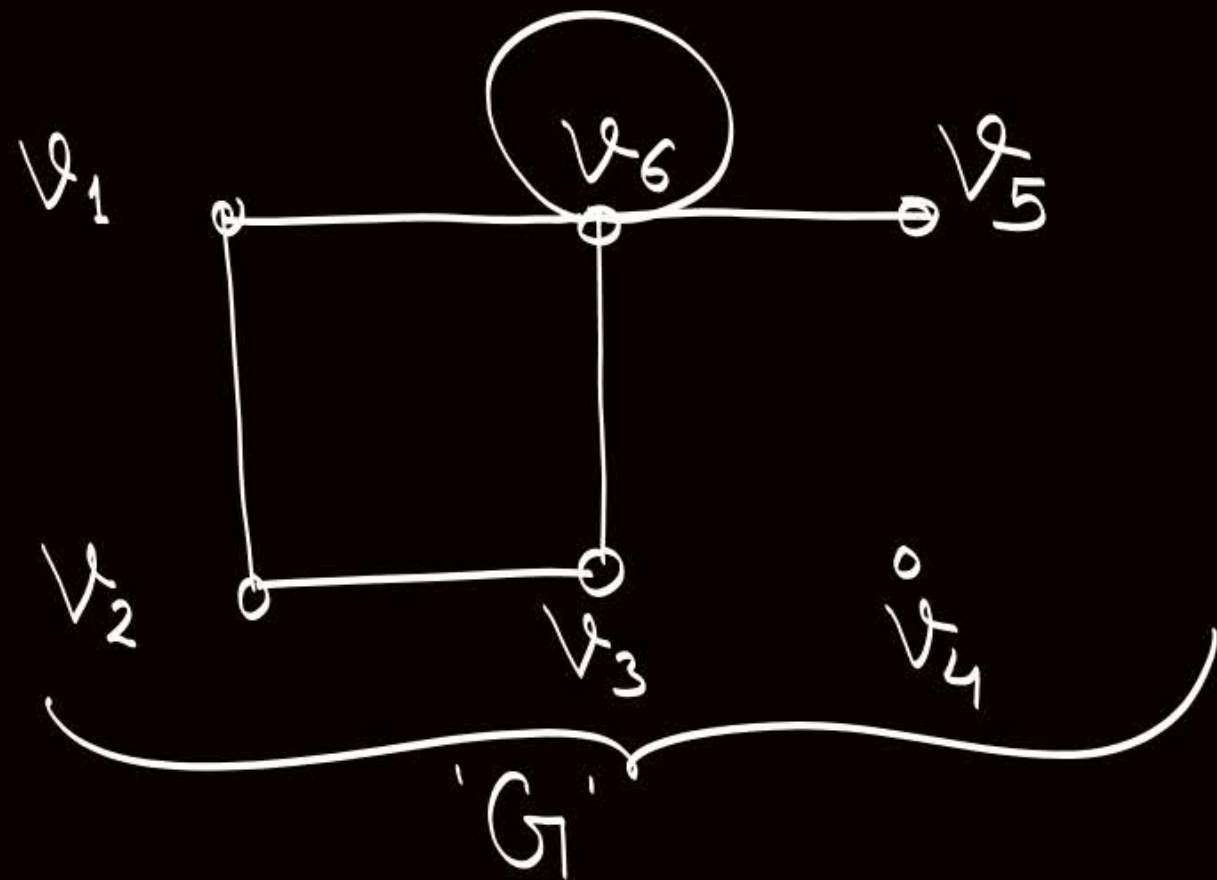


Topic : $\delta(G)$ and $\Delta(G)$

In a graph G

$\delta(G)$: Minimum of the degrees of all the vertices of graph G is denoted by $\delta(G)$

$\Delta(G)$ = Maximum of the degrees of all the vertices of graph G is denoted by $\Delta(G)$



$$\delta(G) = 0$$

$$\Delta(G) = 5$$

$$\deg(v_1) = 2$$

$$\deg(v_2) = 2$$

$$\deg(v_3) = 2$$

$$\deg(v_4) = 0 \Rightarrow "v_4" \text{ is an isolated vertex}$$

$$\deg(v_5) = 1$$

$$\deg(v_6) = 3 + 2 = 5$$



Topic : Null Graph



Vertices may be present

→ A graph with no edges in it is called a NULL graph

Empty Graph: - A graph with no vertex and no edges is called an Empty graph



Topic : Isolated Vertex

In a graph G a vertex of degree $= 0$ is called
an isolated vertex

i.e., if $\deg(v) = 0$, then ' v ' is an isolated vertex



Topic : Regular Graphs

In a graph G , if degree of each vertex is Exactly same, then graph G is called a regular graph

In a graph G , if degree of each vertex is exactly ' k ' then graph G is called a ' k -regular' graph



Every Polygon is
a ' 2 -regular'
graph



Topic : Types of graphs

- ✓ 1. Complete Graph
- ✓ 2. Cycle Graph
- ✓ 3. Wheel Graph
- ✓ 4. Connected Graph
- ✓ 5. Cyclic Graph
- ✓ 6. Acyclic Graph
- ✓ 7. Tree
- ✓ 8. Bipartite Graph
- ✓ 9. Complete Bipartite Graph

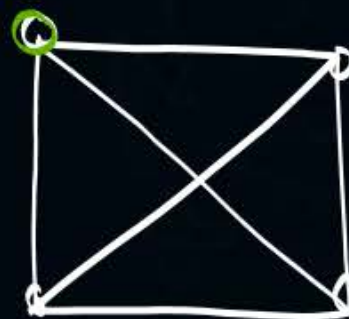


Topic : Complete Graph

K_n

- A complete graph with n -vertices is denoted by ' K_n '
- K_n is a simple graph with n -vertices and all $n_c_2 = \frac{n(n-1)}{2}$ edges

K_1



K_4



K_5

* In a complete graph K_n all n vertices are mutually adjacent

→ In a complete graph K_n degree of each vertex is exactly ' $n-1$ ', ∴ K_n is a ' $n-1$ ' regular graph

• In a complete graph K_n

$$|E(K_n)| = nC_2 = \frac{n(n-1)}{2}$$

↗
No. of Edges in
Complete graph K_n



Topic : Cycle Graph



" C_n "

- ★ A simple graph with n -vertices and n -edges ($n \geq 3$) where all n vertices form a cycle of length ' n ' is called a Cycle graph
- ★ A cycle graph with n -vertices is denoted by ' C_n '
- ★ Length of a cycle = No. of edges forming that cycle



C_3

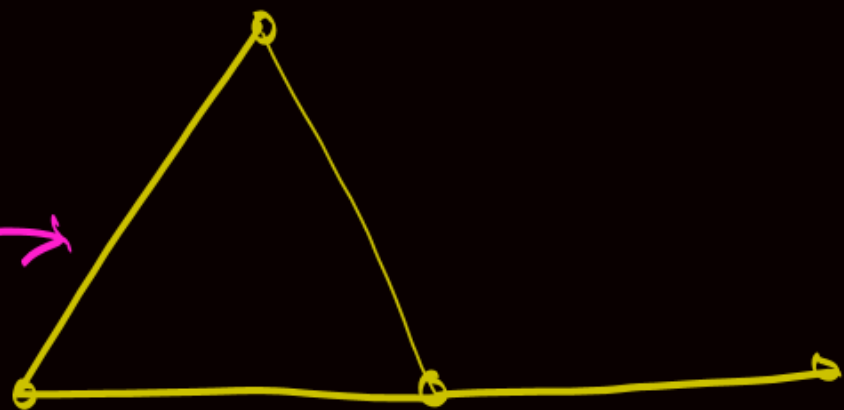


C_4



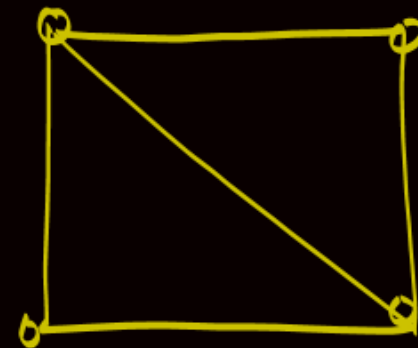
C_5

In a cycle graph C_n
 $|E(C_n)| = n$



'4' vertices
and '4' edges
but is not a Cycle graph

length of cycle = $3 \neq 4$



Not a cycle graph



Topic : Wheel Graph



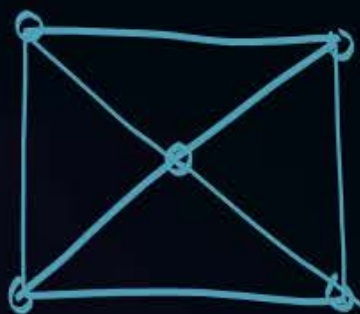
' W_n '

A wheel graph with n -vertices is denoted by W_n .

A wheel graph with n -vertices can be obtained by connecting a hub vertex with all the vertices of cycle graph C_{n-1} .



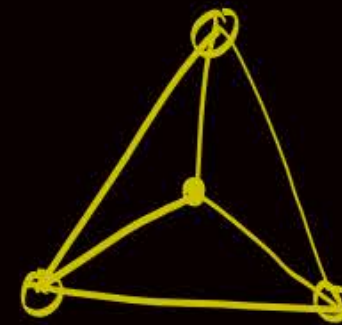
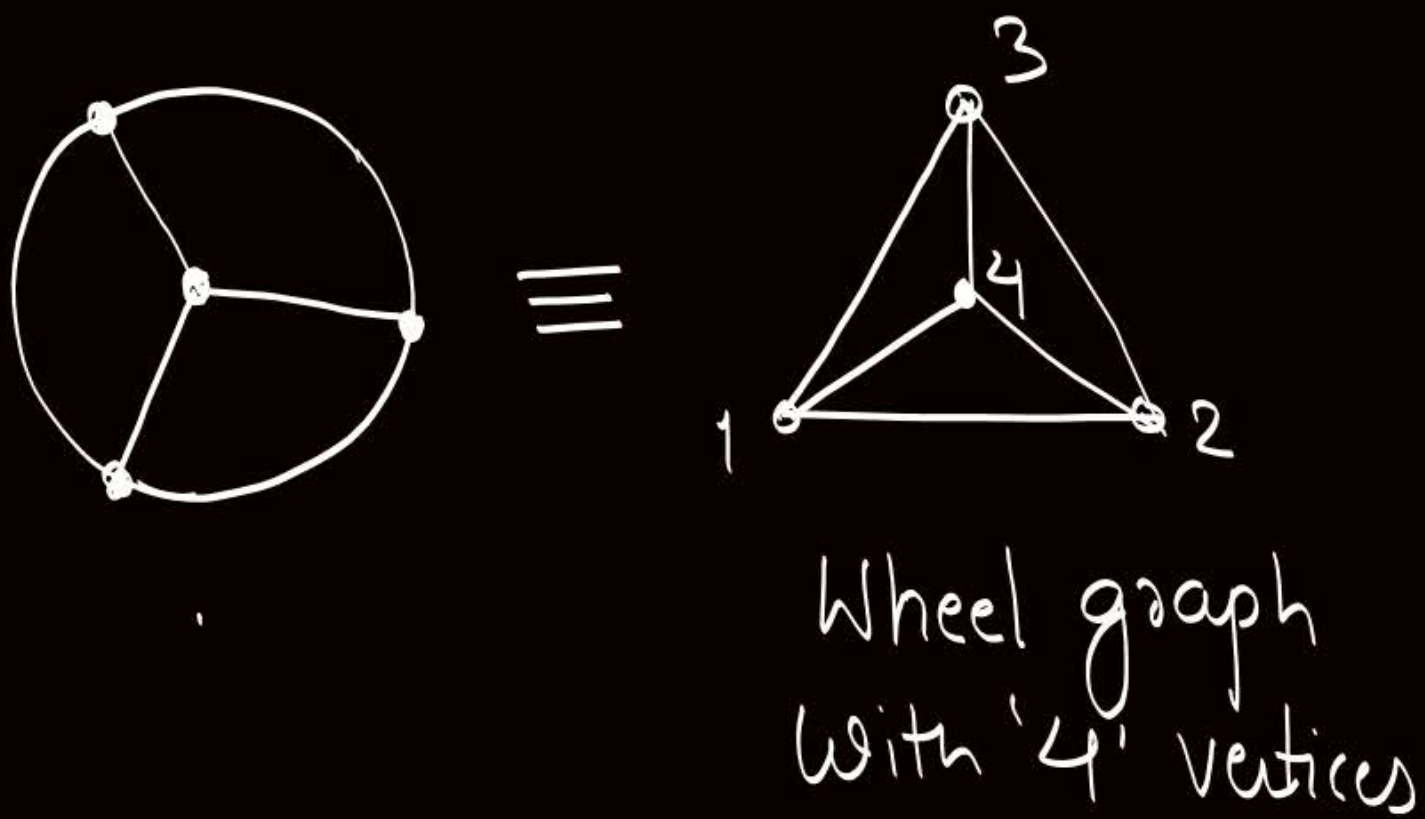
$$W_4 = C_3 + 1\text{-hub}$$



$$W_5 = C_4 + 1\text{-hub}$$



$$W_6 = C_5 + 1\text{-hub}$$



$$W_4 = C_{4-1} + 1\text{-hub}$$

In a wheel graph W_n

$$|E(W_n)| = \underbrace{(n-1)}_{\text{w.r.t. } C_{n-1}} + \underbrace{(n-1)}_{\text{to connect hub vertex with 'n-1' vertices of } C_{n-1}} = 2n-2$$

w.r.t. C_{n-1}

to connect hub vertex with 'n-1' vertices of C_{n-1}

In a wheel graph W_n

- degree of hub vertex = $(n-1)$
- degree of Every other vertex = 3
(Except hub vertex)

Every wheel graph W_n
need not be a
regular graph

- A wheel graph W_n is a regular graph
if and only if

$$n-1 = 3$$
$$\boxed{n=4}$$

The only wheel graph which
is also a regular graph
is W_4

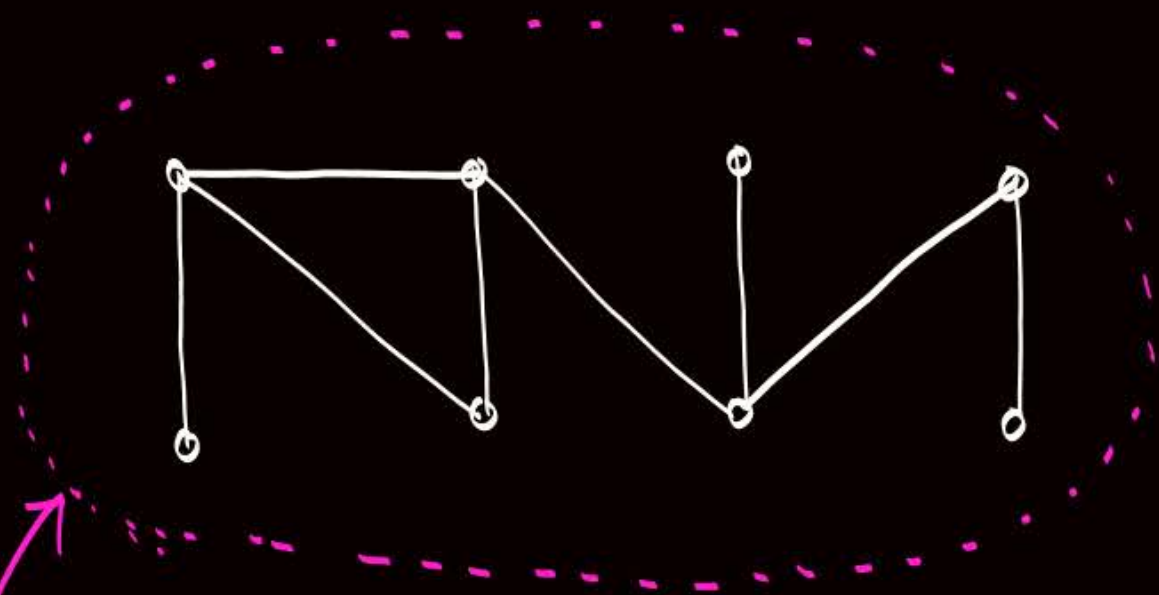


Topic : Connected Graph

→ A graph with two or more vertices is said to be connected if and only if there exist a path between every pair of vertices.

A single isolated vertex may be treated differently in different scenarios.

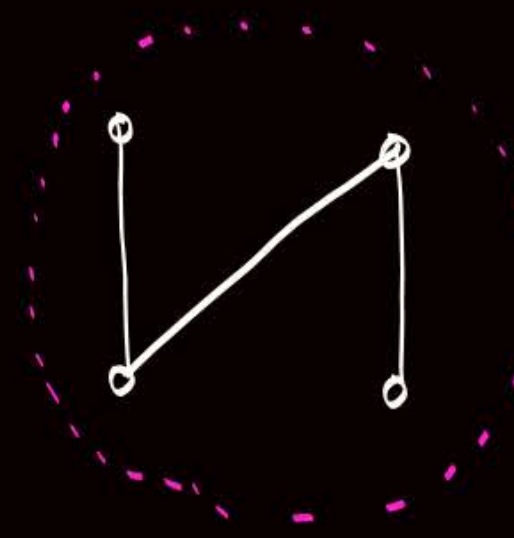
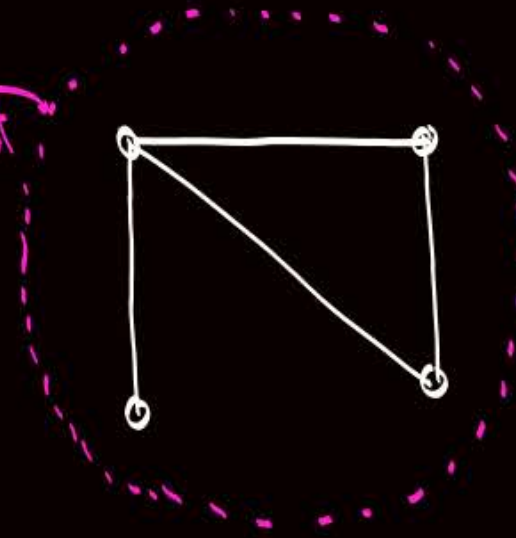
eg:



graph ' G_1 '

In ' G_1 ' there is a single connected component
' G_1 ' is a connected graph + simple

it is a connected component



it is also a connected component

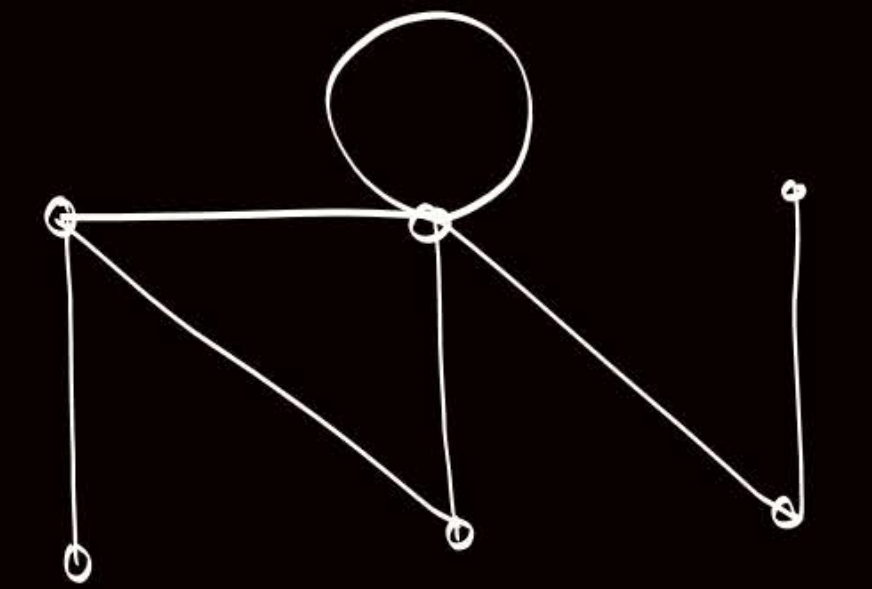
graph ' G_2 '

' G_2 ' is not a connected graph (but simple)
{ In ' G_2 ' more than one }
connected components

→ If graph is a Connected graph then there will be
Exactly '1' Connected Component

→ If graph is a disconnected graph then it will
have two or more Connected Components.

→



It is a Connected graph
but it is not a simple graph

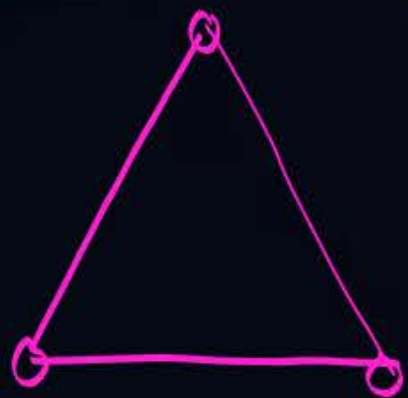
→ A Connected graph
may or may not
be a simple graph

* K_n , C_n and W_n are always simple
as well as connected graph

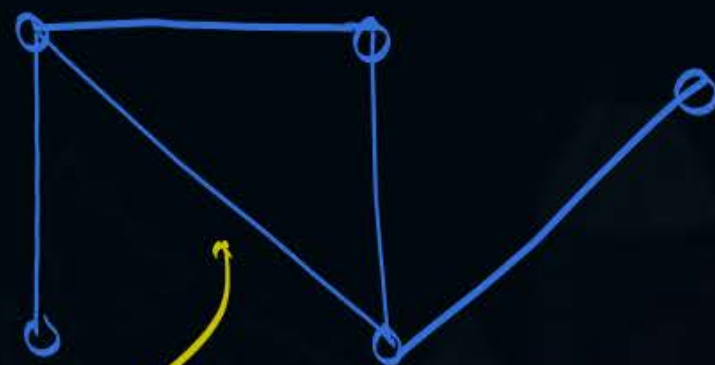


Topic : Cyclic Graph

A graph with at least one cycle in it
is called a cyclic graph

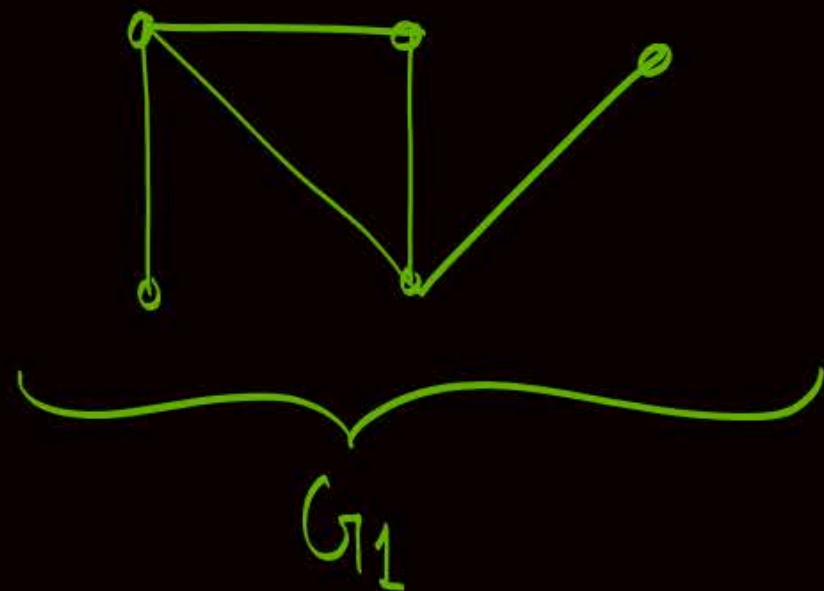


It is a cycle graph
as well as a
cyclic graph

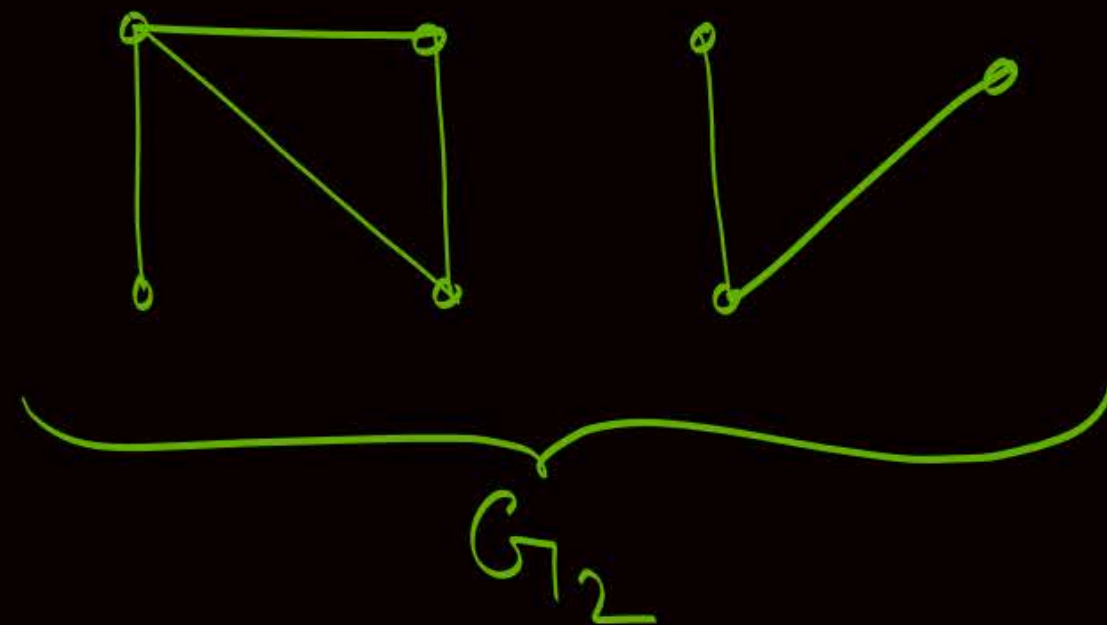


We have at least one cycle in it
∴ it is a cyclic graph,
but it is not a cycle graph

* Every cycle graph is also a cyclic graph,
but Every cyclic graph need not be a cycle graph



Cyclic + graph
+
Connected graph



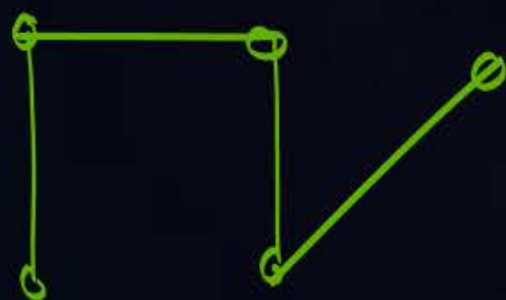
Cyclic + graph
+
disconnected graph

- * A cyclic graph may or may not be connected
- * A cyclic graph may or may not be simple

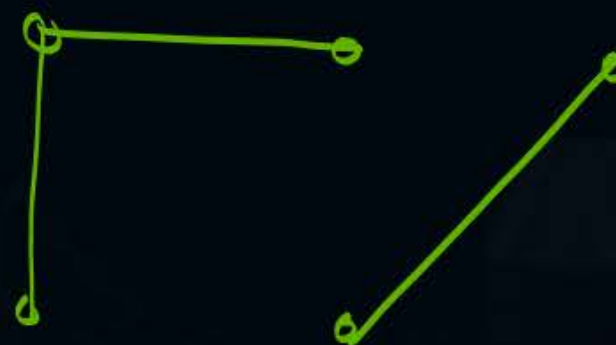


Topic : Acyclic Graph

A graph with no cycle in it is called an 'acyclic' graph.



Acyclic
+
Connected



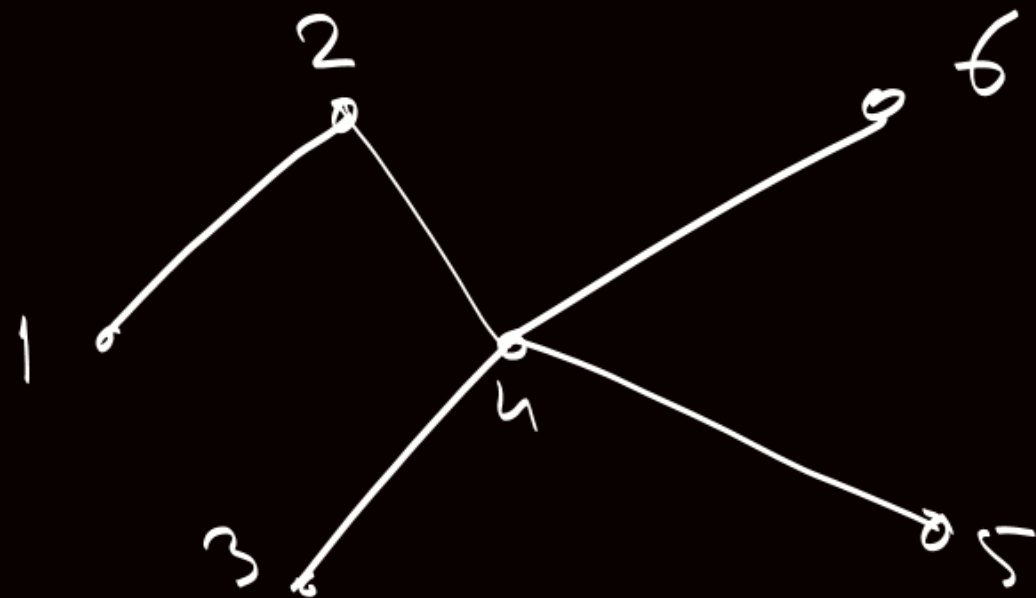
Acyclic
+
disconnected



Topic : Tree



- A tree is a connected and acyclic graph
- A tree with n -vertices will have exactly ' $n-1$ ' edges
- A collection of tree is called a forest
ie. A collection of acyclic components
is called a forest



A tree with
'6' vertices
and '5' edges

- * In a simple graph with n -vertices, if number of edges are more than ' $n-1$ ' {i.e., $|E| > n-1$ }, then graph is guaranteed to be a Cyclic graph {i.e., it can never be a tree}
- * A graph with n -vertices and ' $n-1$ ' edges may or may not be a tree, { it may not be Connected } and { may not be Cyclic }
- * A graph with n -vertices and less than ' $n-1$ ' edges can never be a Connected graph { it may or may not be Cyclic }
- * In a graph with n -vertices if number of edges are more than or equal to $n-1$ {i.e., $|E| \geq n-1$ }, then also it may or may not be Connected

H.W.

Q: A forest with n -vertices and $\underbrace{k \text{ components}}_{\text{i.e., 'k' trees}}$ will have how many edges?



Topic : Bi-partite Graph

ie. No two vertices of the same subset of the partition should be adjacent to each other

- * A bipartite graph (or bigraph) is a graph whose vertices can be partitioned into two sets V_1 and V_2 such that every edge of the graph connects a vertex in set V_1 to a vertex in set V_2 .

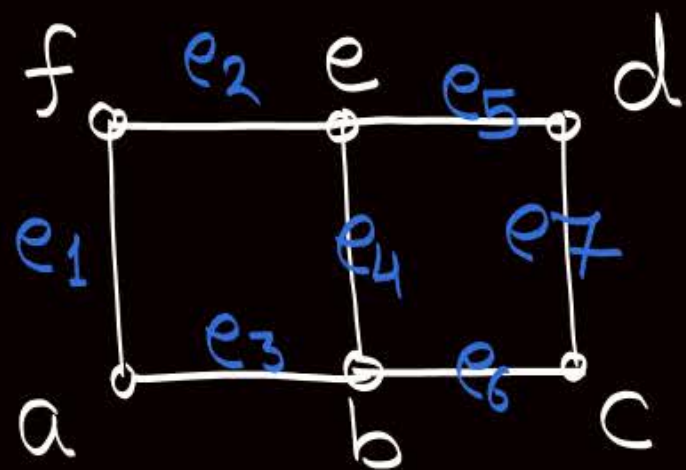
Let $G = (V, E)$

↳ if $V = \{V_1, V_2\}$ is a partition, then $V_1 \cap V_2 = \emptyset$ and $V_1 \cup V_2 = V$

- * A bipartite graph can be represented using the notation

Ordered triple $G = (V_1, V_2, E)$ where,
 V_1 is the first set of the partition
 V_2 is the second set of the partition and
 E is the set of edges of graph G

eg:

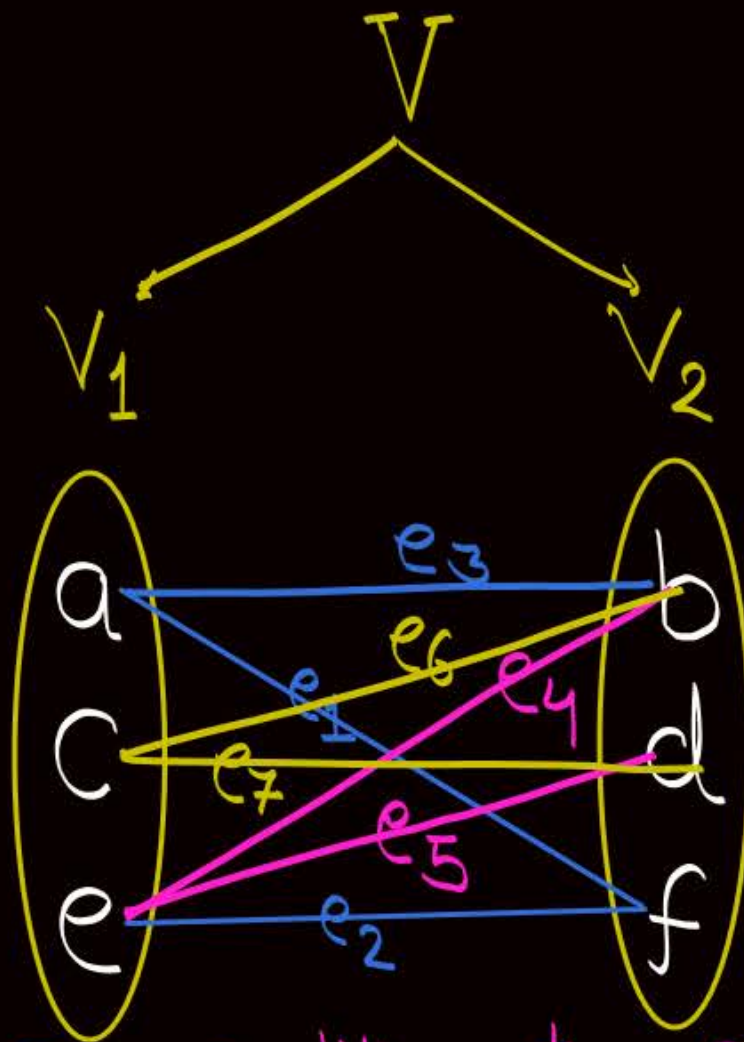


$$G = (V, E)$$

$$V = \{a, b, c, d, e, f\}$$

$$E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$$

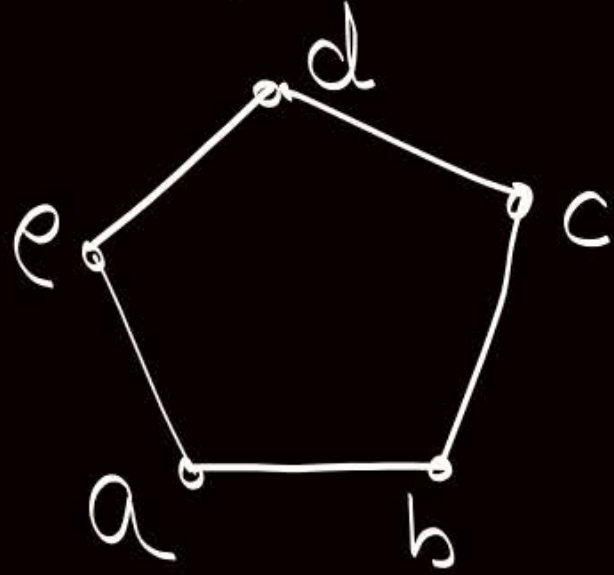
Check whether graph G is a bi-partite graph or not



We can partition the set of vertices into two subsets, s.t. no two vertices of same subset are adjacent to each other, $\therefore G = (V, E)$ is a bipartite graph

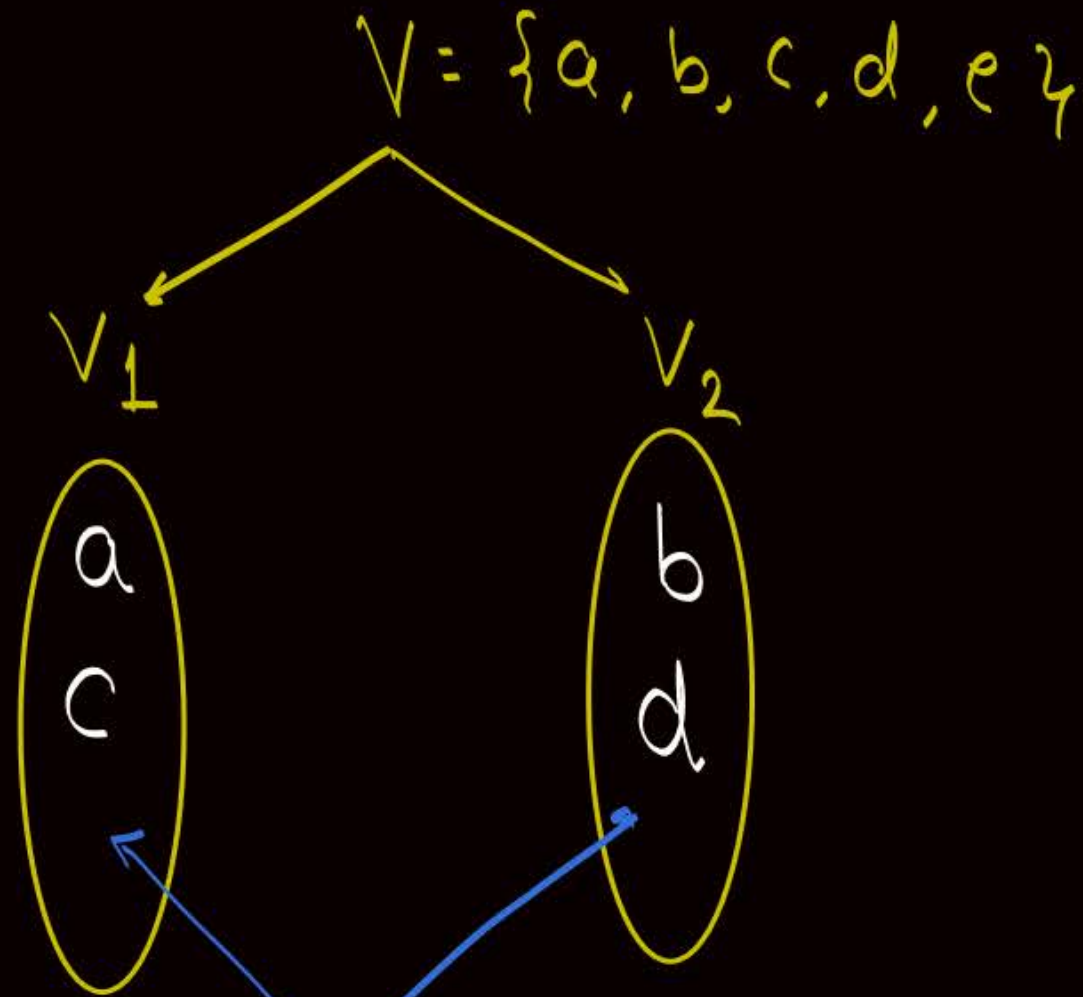
And partition of $V = \{V_1, V_2\}$
s.t. $V_1 = \{a, c, e\}$
 $V_2 = \{b, d, f\}$

Q: Check whether the following graph is a bi-partite graph or not.



it is a cycle of odd length

the following graph is a bi-partite



'e' can not be placed in any of the subset, and no other partition is possible as well. $\therefore$ Graph is not a bipartite graph



2 mins Summary



✓
Topic

Terminologies related to graphs

✓
Topic

Different types of graphs

THANK - YOU